Tacory Bey
Project 1
Computer Networking

The objective of this project was to build and deploy a publicly accessible website while demonstrating fundamental networking concepts used in the real world web infrastructure. Modern websites rely on several networking technologies working together, including Domain Name System (DNS), IP addressing, communication protocols (HTTP/HTTPS), and security mechanisms that protect data during transmission.

For this project, I developed a simple website using HTML and CSS and deployed using Github pages. Github Pages is a hosting service that allows users to publish static websites directly from a Github repo. This hosting method provides a publicly accessible domain and automatically supports secure connections through HTTPS.

The project demonstrates how a domain name resolves to an IP address through DNS, how network communication occurs using TCP/IP protocols, and how secure communication is established through HTTPS encryption. By deploying the website and examining the network interactions using command line tools and browser developer tools, it is possible to observe how these networking components function together in a real deployment environment.

The first step I took when building this website is using Github Pages. The platform integrates directly with Github repository , allowing files stored in a repository to be served as a public website. I created three files for github to use in its pages: index.html, networking.html. and style.css. Index.html was created to be viewed as the front page, on the front page displays class information, project information, as well as a link to the code on github. Networking.html was created to explain the networking concepts such as DNS, IP addressing, protocols, security features, etc. Refer to Image 1 for reference
https://github.com/corybey/CN-Proj1
https://corybey.github.io/CN-Proj1/index.html

DNS is a critical component of the internet that translates human readable domain names into numerical IP addresses. Since computers communicate with IP addresses, DNS allows users to access websites using easier to remember domain names. To demonstrate, I used “nslookup corybey.github.io”. The command sends a DNS request and returns the IP address associated with the domain. The output confirms that the domain resolves correctly and identifies the server responsible for hosting a website. This demonstrates how DNS acts as the translation layer between domain names and IP addresses on the internet. Refer to Image 2 for reference

IP addressing is used to uniquely identify devices connected to a network. Every server and client device on the internet communicates using IP addresses within the TCP/IP networking model. Once DNS resolves a domain name, the browser uses the returned IP to send requests to the appropriate server. To demonstrate, the ping command “ping corybey.github.io” was used. The ping command sends ICMP echo request packets to the server and measures the response time. If the server responds, it confirms that the client device can reach the host through the network. Refer to Image 3 for reference

HTTP is the protocol used by web browsers and servers to exchange web content. However, standard HTTP communication transmits data in plaintext, making it vulnerable to interception or tampering. HTTPS, a different version of HTTP, improves security by encrypting communication using Transport Later Security. Encryption ensures that the data exchanged between the client and server remains confidential and cannot easily be intercepted or modified. During my audit, I checked what protocol was being used, the status code for those items loaded, and a secure lock icon for the browser.

The security feature implemented in this project is HTTPS enforcement. Github Pages automatically enables HTTPS encryption for hosted websites and allows administrators to enforce secure connections. HTTPS protects communication between the client and server by encrypting transmitted data. This prevents attackers from intercepting sensitive information and any potential modification of data.

During the deployment process, an initial issue occurred where the website did not appear immediately after enabling Github Pages. This happened because Github Pages specifically wants index.html as the root file in the directory. Without it, github cannot determine the default page.

In conclusion, this project demonstrated how a public website can be deployed while applying core networking concepts. Through the use of Github pages, the website became publicly accessible through a domain that resolves via DNS to an IP hosted by Github. The project also illustrated how IP addressing allows client devices to communicate with web servers and how HTTPS ensures secure communication through encryption. In summary, the final deployed site demonstrates that the website is publicly hosted, DNS resolves the domain to an IP address, and communication occurs over HTTPS using TCP/IP protocols.

Image 1

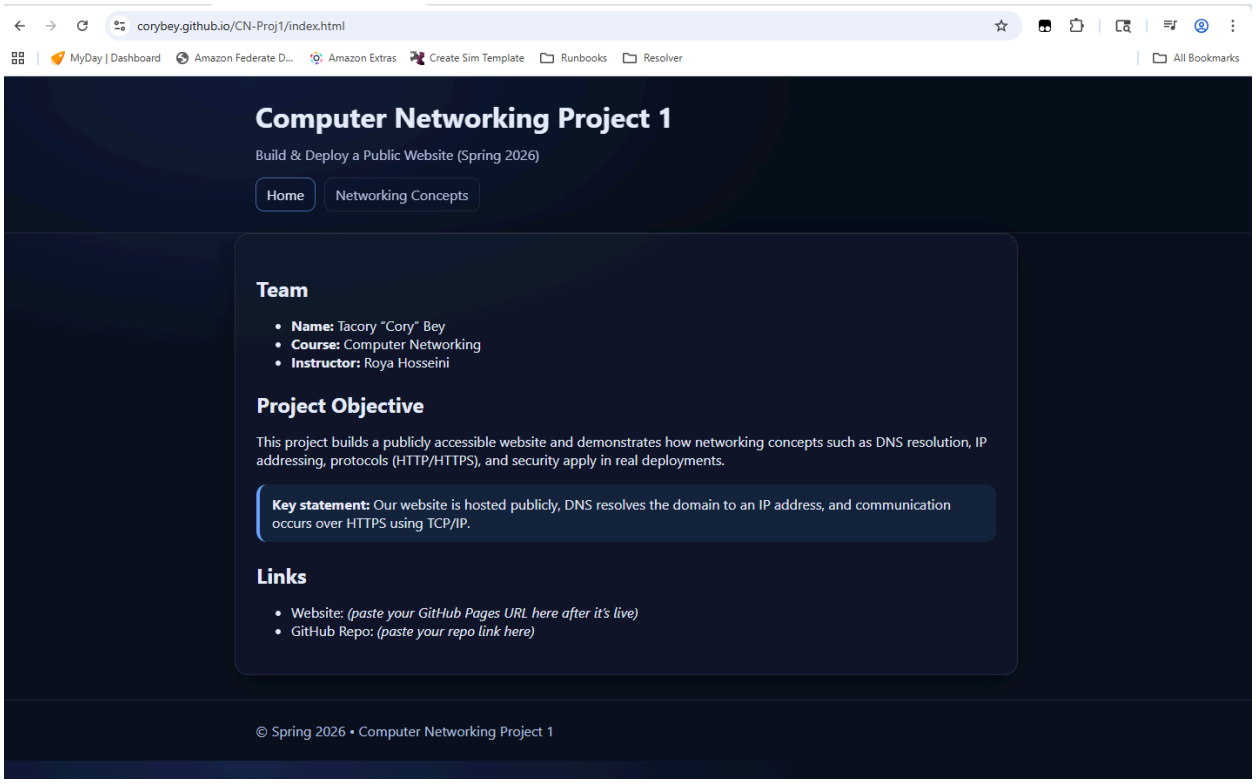
corybey.github.io/CN-Proj1/index.html
MyDay | Dashboard
Amazon Federate D...
Amazon Extras
Create Sim Template
Runbooks
Resolver
All Bookmarks
Computer Networking Project 1
Build & Deploy a Public Website (Spring 2026)
Home
Networking Concepts
Team
Name: Tacory "Cory" Bey
Course: Computer Networking
Instructor: Roya Hosseini
Project Objective
This project builds a publicly accessible website and demonstrates how networking concepts such as DNS resolution, IP addressing, protocols (HTTP/HTTPS), and security apply in real deployments.
Key statement: Our website is hosted publicly, DNS resolves the domain to an IP address, and communication occurs over HTTPS using TCP/IP.
Links
Website: (paste your GitHub Pages URL here after it's live)
GitHub Repo: (paste your repo link here)
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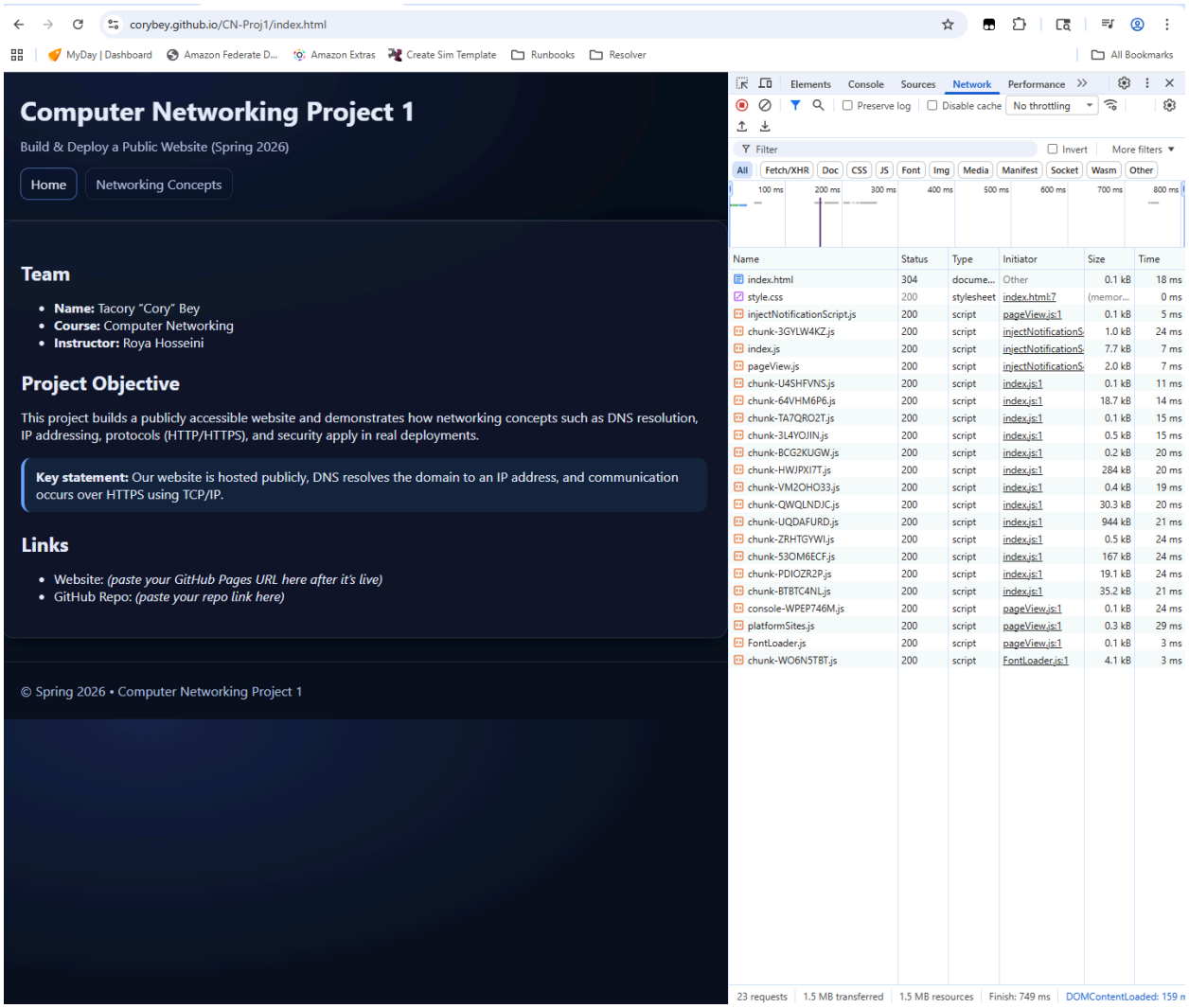
Image 2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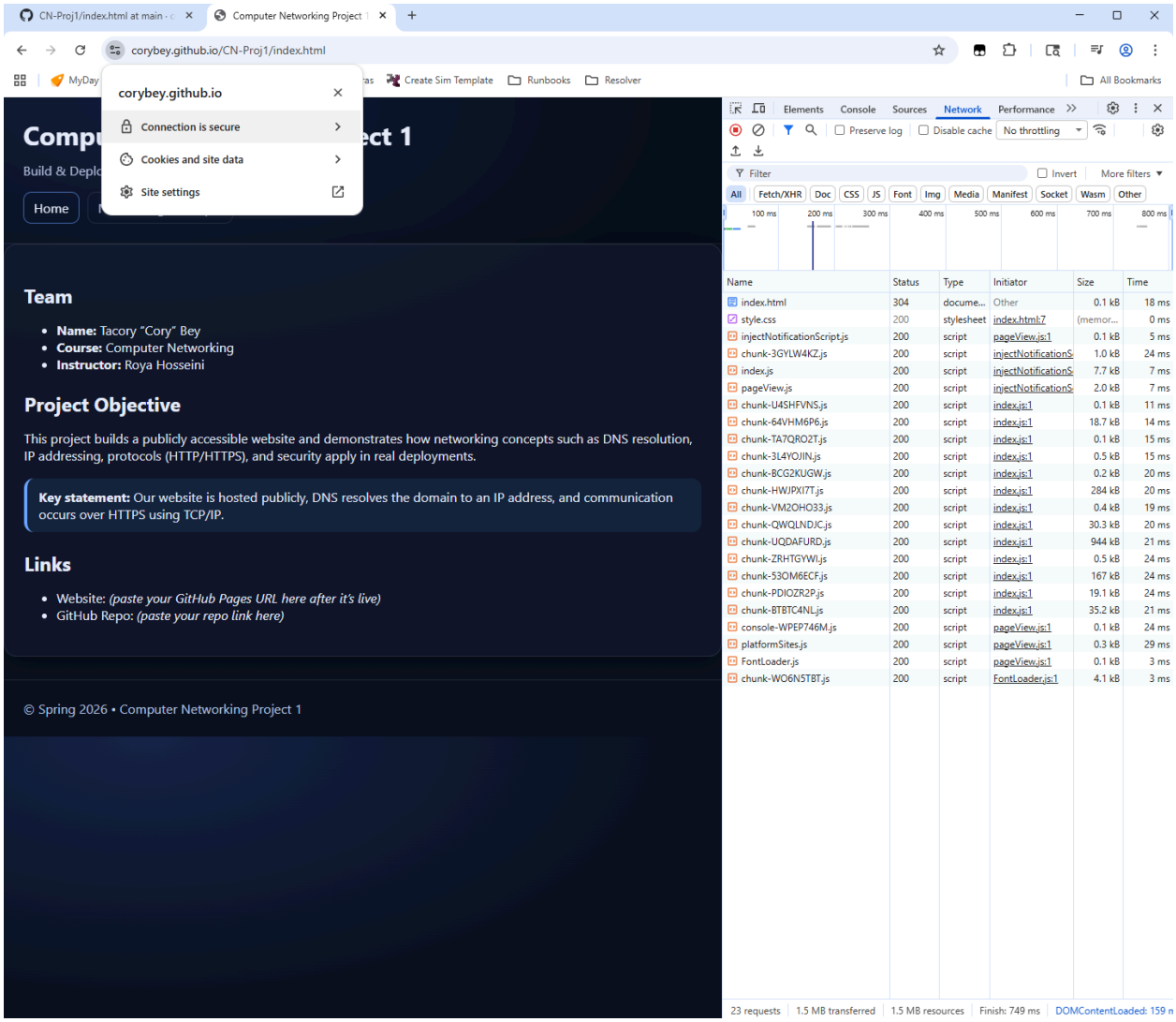
Disable cache
No throttling
Filter
Invert
More filters
All
Fetch/XHR
Doc
CSS
JS
Font
Img
Media
Manifest
Socket
Wasm
Other
100 ms
200 ms
300 ms
400 ms
500 ms
600 ms
700 ms
800 ms
Name Status Type Initiator Size Time
index.html 304 docume... Other 0.1 kB 18 ms
style.css 200 stylesheet index.html:7 (memor... 0 ms
injectNotificationScript.js 200 script pageView.js:1 0.1 kB 5 ms
chunk-3GYLW4KZ.js 200 script injectNotificationS 1.0 kB 24 ms
index.js 200 script injectNotificationS 7.7 kB 7 ms
pageView.js 200 script injectNotificationS 2.0 kB 7 ms
chunk-U4SHFVNS.js 200 script index.js:1 0.1 kB 11 ms
chunk-64VHM6P6.js 200 script index.js:1 18.7 kB 14 ms
chunk-TA7QRO2T.js 200 script index.js:1 0.1 kB 15 ms
chunk-3L4YOJIN.js 200 script index.js:1 0.5 kB 15 ms
chunk-BCG2KUGW.js 200 script index.js:1 0.2 kB 20 ms
chunk-HWJPXI7T.js 200 script index.js:1 284 kB 20 ms
chunk-VM2OHO33.js 200 script index.js:1 0.4 kB 19 ms
chunk-QWQLNDJC.js 200 script index.js:1 30.3 kB 20 ms
chunk-UQDAFURD.js 200 script index.js:1 944 kB 21 ms
chunk-ZRHTGYWI.js 200 script index.js:1 0.5 kB 24 ms
chunk-53OM6ECF.js 200 script index.js:1 167 kB 24 ms
chunk-PDIOZR2P.js 200 script index.js:1 19.1 kB 24 ms
chunk-BTBTC4NL.js 200 script index.js:1 35.2 kB 21 ms
console-WPEP746M.js 200 script pageView.js:1 0.1 kB 24 ms
platformSites.js 200 script pageView.js:1 0.3 kB 29 ms
FontLoader.js 200 script pageView.js:1 0.1 kB 3 ms
chunk-WO6N5TBT.js 200 script FontLoader.js:1 4.1 kB 3 ms
23 requests
1.5 MB transferred
1.5 MB resources
Finish: 749 ms
DOMContentLoaded: 159 n

Image 3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JS
Font
Img
Media
Manifest
Socket
Wasm
Other
Name
Status
Type
Initiator
Size
Time
index.html 304 docume... Other 0.1 kB 18 ms
style.css 200 stylesheet index.html:7 (memor... 0 ms
injectNotificationScript.js 200 script pageView.js:1 0.1 kB 5 ms
chunk-3GYLW4KZ.js 200 script injectNotificationS 1.0 kB 24 ms
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